



Design and kinematic analysis of a 3-RRR spherical parallel manipulator reconfigured with four-bar linkages

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ABSTRACT

In this paper, the 3-RRR spherical parallel manipulator (SPM) of co-axis input is redesigned with reconfigurability. By utilizing a simple four-bar linkage, the SPM is able to change virtually the dimensions of driving links. The new design allows the SPM reconfigurable for investigation of the influence of the link dimensions on the kinematic and dynamic performance of the SPM. Moreover, the new design can be used directly for the applications such as flight simulator. In this paper, the workspace, dexterity and singularity of the new design are investigated, considering the effect of an adjustable link dimension onto the manipulator performance. The relationship between the varying link dimension in a prescribed range and the operational workspace is studied, with some new results on workspace and dexterity of the SPM revealed.

1. Introduction

Three-degree-of-freedom (3-dof) spherical parallel manipulators (SPMs) have been studied for several decades. Typical SPMs are 3-RRR type manipulators [1–5], where R stands for the revolute joint and an underlined letter stands for an actuated joint, as shown in Fig. 1(a), such as the ‘Agile Eye’ [6]. Designs with prismatic joints have been presented, for instance in [7–12].

The exiting SPM architectures usually lead to the limited ranges of rotational motions, particularly, the twist movement. On the other hand, some applications demand a larger twist angle, such as the active spherical joints. For this purpose, a co-axial input SPM with full-circle twist was proposed [5,13], with three actuation units sliding on a circular guide, as depicted in Fig. 1(b). Kinematically, the sliding body can be treated as a special case of revolute joint of 90 degrees, analogy to planar prismatic joint special case of revolute joint centered as infinity. Compared to Fig. 1(a), this manipulator is a special case of $\gamma = 0$, where all three joints on the base are co-axial. Recently, an integrated design combining both SPM and magnetic actuation was reported in [14]. The design optimization of this SPM carried out in [5,15] showed that the angle α_1 significantly influences the manipulator performance. Thus, the primary objective of this paper is to study further how this dimension can be physically adjusted and the influences of adjustment to the kinematic performance. This is achieved by means of reconfiguration.

Reconfigurable manipulators were developed as a scalable and flexible solution for modern automation and fine manipulation systems [16], by gathering the advantages of several robotic architectures in a single system. Aiming to change mobilities [17–19], to avoid singular poses [20], to enlarge the workspace [16,21] or to change its dynamic properties [22–25], reconfiguration can be achieved statically or dynamically [26,27]. Static reconfiguration requires physical intervention from the user, to add or remove links or connecting elements to the structure [16,17,21,25]. By contrast, dynamic reconfiguration is much more flexible as it can realize online reconfiguration, but with increased complexity, which has been reported in the literature too. A 6-dof reconfigurable hybrid parallel manipulator [22] adopts variable length joints on its base for the reconfiguration. Reconfigurable hook joints are adopted in a 3-dof parallel mechanism to change the directions of the axes of rotation in the limbs to produce translations or rotations [28]. Reconfigurable parallel manipulators also use adaptive joints to block one or more DOF in the operational mode [29–31]. An electromagnetic brake clutch was adopted in the NaVARo manipulator [32] to allow the latter to switch among its eight assembly modes. Reconfiguration has proven to be a solid strategy for the improvement of flexibilities of parallel robotics.

In this work, reconfiguration approach is applied to the SPM in Fig. 1(b) to change the workspace size and dexterity. As shown in Fig. 2, the solution is to use a kinematic linkage to change the angle α_1 . Moreover, the moving platform of the manipulator can obtain an

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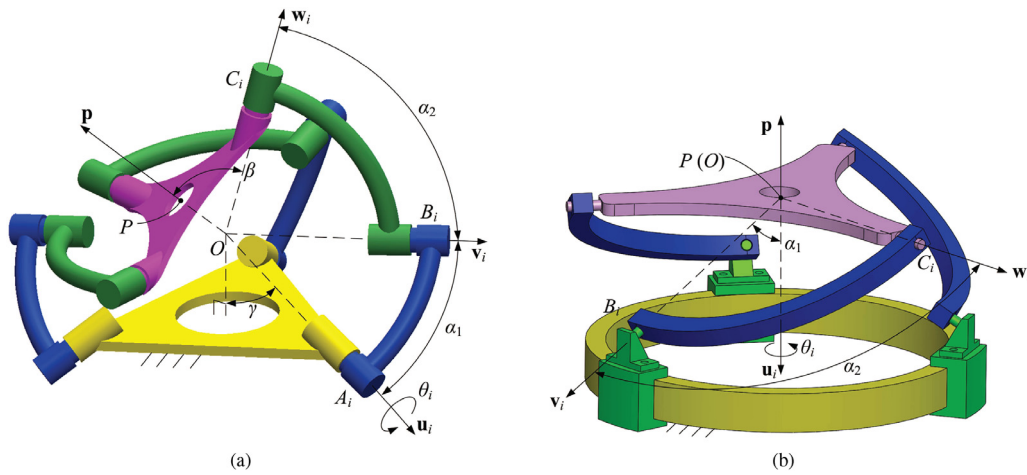


Fig. 1. Architecture of 3-RRR SPM: (a) general, (b) co-axis input.

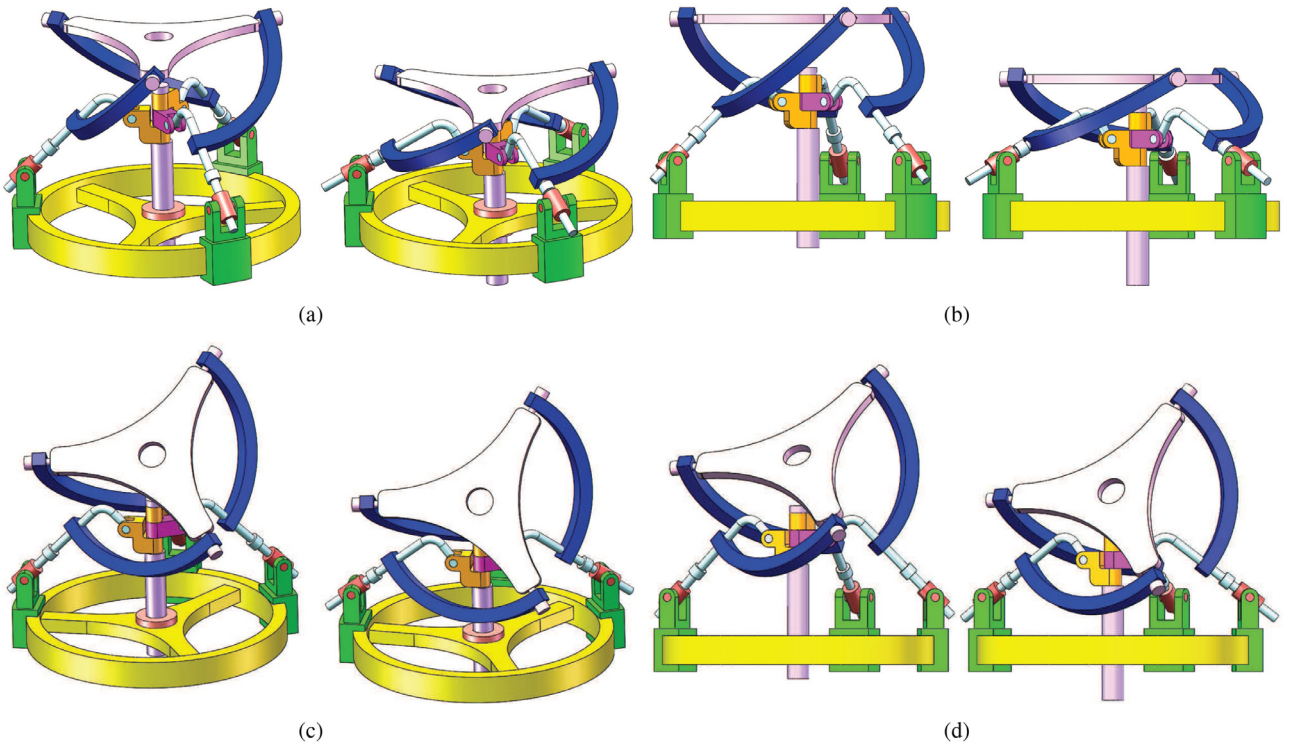


Fig. 2. The wrist manipulator at different configurations: (a) isometric view and (b) front view with sliders evenly distributed; (c) isometric view and (d) front view with sliders arbitrarily distributed.

additional translation along the vertical axis during its reconfiguration, which implies that the mechanism can generate 3R1T motion and function as a flight simulator. In this paper, the workspace, dexterity and singularity of the reconfigurable SPM will be studied to reveal the kinematic characteristics of this new manipulator.

2. Reconfigurable SPM architecture

A rendered CAD model of the reconfigurable SPM is depicted in Fig. 3(a), of which the curved links are driven by actuators moving independently on a circular guide via pinion and gear-ring transmissions. A set of circular guide of model HCR from THK company is used to enable sliders' high-precision circular motion with small clearance. Hard springs and elastic washers are arranged at both sides of the holes on the curved links to reduce the clearance. The actuated linear motion of the linkage along the vertical direction is realized via a driven ball-

screw system, as the ball-screw actuation can transform a small torque into a larger force with a low-power actuator, which is preferable in this design, compared to the linear actuator. Moreover, the sleeve bearings are adopted for structural compactness. The simplified CAD model of the reconfigurable SPM is shown in Fig. 3(b). A $\underline{C}\underline{R}\underline{P}\underline{R}$ (P and C denote prismatic and cylindrical joints, respectively.) linkage is added to the SPM in Fig. 1(b) in each limb, which can be considered as a $\underline{P}\underline{R}\underline{P}\underline{R}$ linkage in plane. The P pairs are driven simultaneously to change the identical angle α_1 .

Fig. 3(b) shows the coordinate system for the kinematic analysis of the manipulator. The i th leg consists of three revolute joints that are parallel to the unit vectors $\mathbf{u}_i = -\mathbf{k}$, $\mathbf{v}_i$, and $\mathbf{w}_i$, $\mathbf{k}$ being the unit vector of z -axis. All three legs have identical architectures, defined by angles α_1 and α_2 , where $\alpha_2 = 90^\circ$ has been proved to be an optimum parameter [15]. The linkage to adjust the angle α_1 lies in the plane determined by vectors $\mathbf{u}_i$ and $\mathbf{v}_i$. Geometric parameters of this linkage are shown in

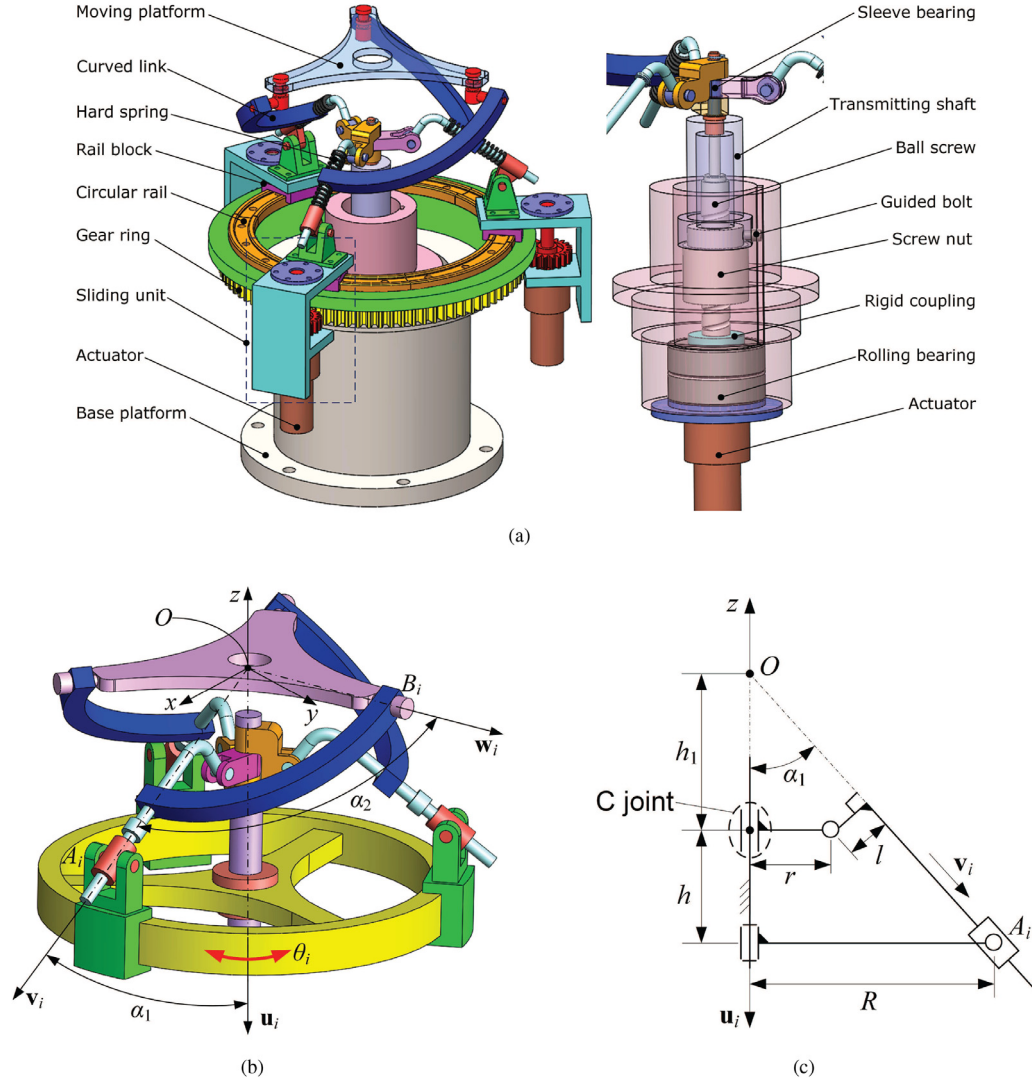


Fig. 3. The reconfigurable SPM: (a) rendered CAD model, (b) parameterization, (c) adjusting linkage, which is essentially a four-bar linkage of PRPR type.

Fig. 3(c).

3. SPM kinematic model

The orientation of the moving platform (MP) is described by an intuitive representation of the *azimuth-tilt-torsion* (ϕ - θ - σ) angles [33] as displayed in Fig. 4, for which the rotation matrix is expressed as:

$$\mathbf{Q} = \mathbf{R}_z(\phi)\mathbf{R}_y(\theta)\mathbf{R}_z(\sigma - \phi) \quad (1)$$

where $\phi \in (-\pi, \pi]$, $\theta \in [0, \pi)$, $\sigma \in (-\pi, \pi]$.

Under the prescribed coordinate system, unit vector $\mathbf{u}_i$ is derived as

$$\mathbf{u}_i = -\mathbf{k} = [0 \ 0 \ -1]^T, \quad i = 1, 2, 3 \quad (2)$$

Unit vector $\mathbf{v}_i$ of the axis of the intermediate revolute joint in the i th leg is expressed as:

$$\mathbf{v}_i = [-\sin(\eta_i - \theta_i)\sin\alpha_i \quad \cos(\eta_i - \theta_i)\sin\alpha_i \quad -\cos\alpha_i]^T \quad (3)$$

where $\eta_i = 2(i-1)\pi/3$, $i = 1, 2, 3$, and $\mathbf{v}_i^*$ represent the unit vectors in the home configuration, namely,

$$\mathbf{v}_i^* = [-\sin\eta_i\sin\alpha_i \quad \cos\eta_i\sin\alpha_i \quad -\cos\alpha_i]^T \quad (4)$$

Unit vector $\mathbf{w}_i$ of the top revolute joint in the i th leg, is a function of the orientation of the moving platform described as

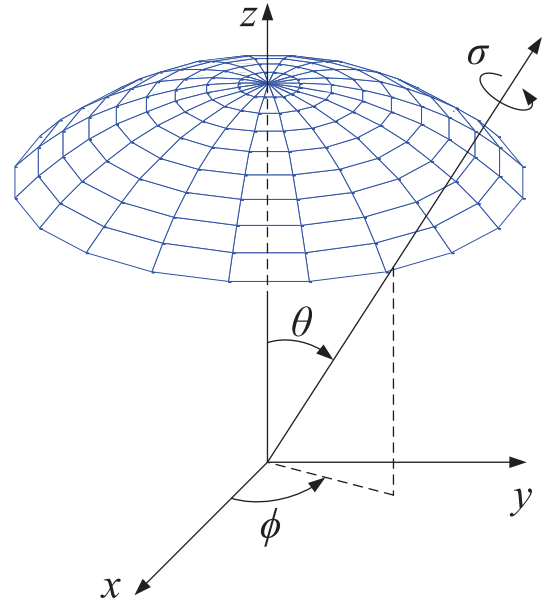


Fig. 4. Orientation representation of the *azimuth-tilt-torsion* angles.

$$\mathbf{w}_i = [w_{ix} \ w_{iy} \ w_{iz}]^T = \mathbf{Q}\mathbf{w}_i^* \quad (5)$$

where $\mathbf{w}_i^*$ is the unit vector for the axis of the top revolute joint of the i th leg when the moving platform (MP) reaches its home configuration, i.e., the MP orientation in the reference frame, which is given as

$$\mathbf{w}_i^* = [-\sin \eta_i \ \cos \eta_i \ 0]^T \quad (6)$$

3.1. Kinematics of the PRPR linkage

While the kinematics of four-bar linkage with prismatic joint(s) has been well-documented in literature [34], we formulate the I/O equation briefly with geometric approach. According to Fig. 3(c), the following geometric relationship exists:

$$\tan \alpha_1 = \frac{R - r - l/\cos \alpha_1}{h} \quad (7)$$

sequentially, the linear displacement along z -axis of the C joint can be expressed as

$$h = \frac{(R - r)\cos \alpha_1 - l}{\sin \alpha_1} \quad (8)$$

If λ denotes the pitch of the ball screw, the input angle θ_z to drive the linkage is calculated as

$$\theta_z = \frac{2\pi h}{\lambda} = \frac{2\pi(R \cos \alpha_1 - r \cos \alpha_1 - l)}{\lambda \sin \alpha_1} \quad (9)$$

3.2. Inverse geometry

The inverse geometry problem has been well documented [35], which can be solved from the following kinematic constraint equation:

$$\mathbf{v}_i \cdot \mathbf{w}_i - \cos \alpha_2 = 0 \quad (10)$$

To this end, the angle θ_i of the i th actuate joint is solved as

$$\theta_i = 2 \tan^{-1} \frac{-I_i \pm \sqrt{I_i^2 + J_i^2 - K_i^2}}{K_i - J_i} \quad (11)$$

with

$$I_i = (w_{ix} \cos \eta_i + w_{iy} \sin \eta_i) \sin \alpha_1 \quad (12a)$$

$$J_i = (-w_{ix} \sin \eta_i + w_{iy} \cos \eta_i) \sin \alpha_1 \quad (12b)$$

$$K_i = -w_{iz} \cos \alpha_1 - \cos \alpha_2 \quad (12c)$$

The inverse geometric problem has eight solutions corresponding to the eight working modes characterized by the sign “−/+” of “ $\pm$ ” in Eq. (11). Here, the working mode with the signs “− −” in Eq. (11) for all the three limbs is selected.

3.3. Kinematic Jacobian matrix

Let $\omega = [\omega_x \ \omega_y \ \omega_z]^T$ denote the angular velocity of the moving platform and $\dot{\theta} = [\dot{\theta}_1 \ \dot{\theta}_2 \ \dot{\theta}_3]^T$ denote the input angle velocities, differentiating Eq. (10) with respect to time leads to

$$\mathbf{A}\omega = \mathbf{B}\dot{\theta} \quad (13)$$

with

$$\mathbf{A} = [\mathbf{a}_1 \ \mathbf{a}_2 \ \mathbf{a}_3]^T; \quad \mathbf{a}_i = \mathbf{v}_i \times \mathbf{w}_i \quad (14a)$$

$$\mathbf{B} = \text{diag}[b_1 \ b_2 \ b_3]; \quad b_i = \mathbf{u}_i \cdot (\mathbf{v}_i \times \mathbf{w}_i) \quad (14b)$$

where matrices $\mathbf{A}$ and $\mathbf{B}$ are named the forward and inverse Jacobian matrices of the manipulator, respectively. The kinematic Jacobian matrix $\mathbf{J}$ of the manipulator [35] is obtained as

$$\mathbf{J} = \mathbf{B}^{-1}\mathbf{A} = [\mathbf{j}_1 \ \mathbf{j}_2 \ \mathbf{j}_3]^T; \quad \mathbf{j}_i = \mathbf{a}_i/b_i \quad (15)$$

4. Workspace, dexterity and singularity analysis

In the development of manipulators, performance evaluation/analysis is necessary to meet design requirements, wherein the workspace (WS) size and quality, reflecting orientational capability, overall kinematic performance indicated by dexterity as well as the singularities, are key issues.

This section is devoted to the kinematic analysis of the aforementioned problems as well as the effect of the adjustable dimensions onto the manipulator performance. Without losing generality, the influence of the adjustable angle α_1 to the kinematic performance will be investigated, sequentially, the link dimension's effect can be readily deduced from either Eq. (8) or Eq. (9) with prescribed link lengths.

4.1. Orientation workspace

As this special SPM can realize the full-circle twist, an efficient approach to find its orientation workspace is to search the workspace with $\sigma = 0$ for axis-symmetric rotation. This workspace is subject to the existing real solutions of inverse geometry problem, hence, the following inequalities can be deduced, namely,

$$f_i(\phi, \theta) = I_i^2 + J_i^2 - K_i^2 \geq 0, \quad i = 1, 2, 3 \quad (16)$$

Substituting the geometric parameters η_i and α_2 into the above equation, the rotation ranges of ϕ and θ can be determined with different angle α_1 from the inequalities below,

$$\sin^2 \alpha_1 - \sin^2 \phi \sin^2 \theta \geq 0 \quad (17a)$$

$$\sin^2 \alpha_1 - \sin^2 \theta \left[\sin \phi \sin \left(\frac{\pi}{3} - \phi \right) + \frac{3}{4} \right] \geq 0 \quad (17b)$$

$$\sin^2 \alpha_1 + \sin^2 \theta \left[\sin \phi \sin \left(\frac{\pi}{3} + \phi \right) - \frac{3}{4} \right] \geq 0 \quad (17c)$$

It is noteworthy that the workspace boundaries of the manipulator, namely, the minimum angle θ , can be detected when the kinematic constraints Eqs. (17a)–(17c) are equal to zero. According to the manipulator geometry, there exists $\alpha_1 \in (0, \pi/2]$, $\theta \in [0, \pi/2]$, thus, the following equations can be obtained:

$$\sin^2 \theta = \frac{\sin^2 \alpha_1}{\sin^2 \phi} \quad (18a)$$

$$\sin^2 \theta = \frac{\sin^2 \alpha_1}{\sin \phi \sin(\pi/3 - \phi) + 3/4} \quad (18b)$$

$$\sin^2 \theta = -\frac{\sin^2 \alpha_1}{\sin \phi \sin(\pi/3 + \phi) - 3/4} \quad (18c)$$

With $\phi \in (-\pi, \pi]$, the following expressions can be derived,

$$\sin \phi \in [-1, 1] \quad (19a)$$

$$\sin \phi \sin \left(\frac{\pi}{3} - \phi \right) + \frac{3}{4} \in [0, 1] \quad (19b)$$

$$\sin \phi \sin \left(\frac{\pi}{3} + \phi \right) - \frac{3}{4} \in [-1, 0] \quad (19c)$$

According to Eqs. (18a)–(18c), the *tilt* angle θ reaches to its minimum when the denominator reaches the maximum, i.e., the absolute values of Eqs. (19a)–(19c) at the left-hand side are equal to 1. Sequentially, with Eqs. (18a)–(18c) separately satisfied, the minimum *tilt* angle θ with different α_1 of all the three limbs can be deduced from $\sin^2 \theta = \sin^2 \alpha_1$. This means that the relationship between the angle α_1 and the regular workspace size (i.e., the minimum θ) is linear, namely, the *tilt* angle θ of the workspace is equal to angle α_1 for $\{\phi, \sigma\} \in (-\pi, \pi]$.

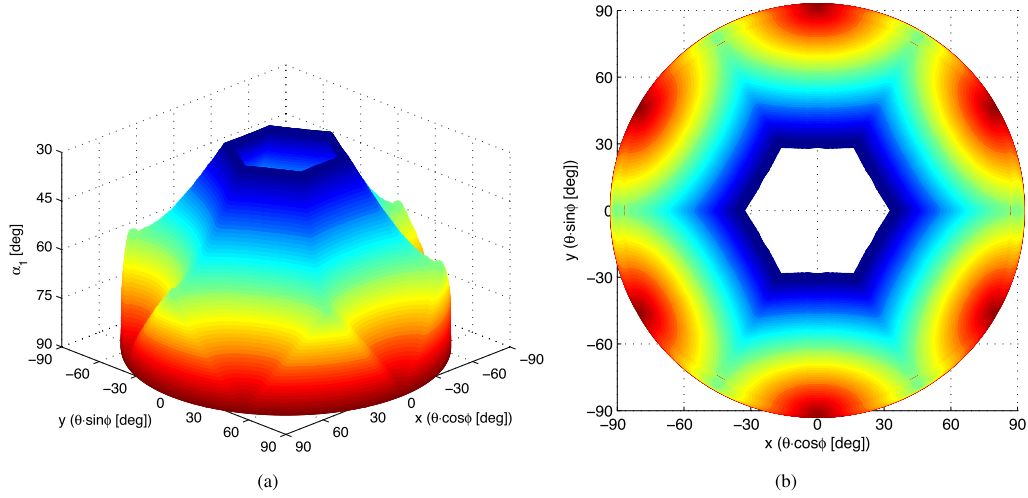


Fig. 5. The reachable zero-twist workspace bounded by 'tower' surface with varying α_1 : (a) isometric view; (b) top view.

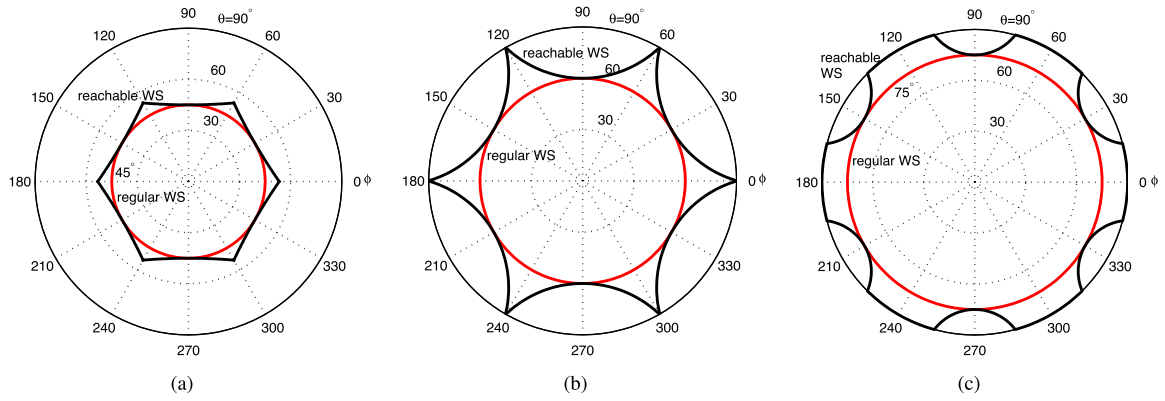


Fig. 6. The zero-twist workspace with two configurations: (a) $\alpha_1 = 45^\circ$; (b) $\alpha_1 = 60^\circ$; (c) $\alpha_1 = 75^\circ$.

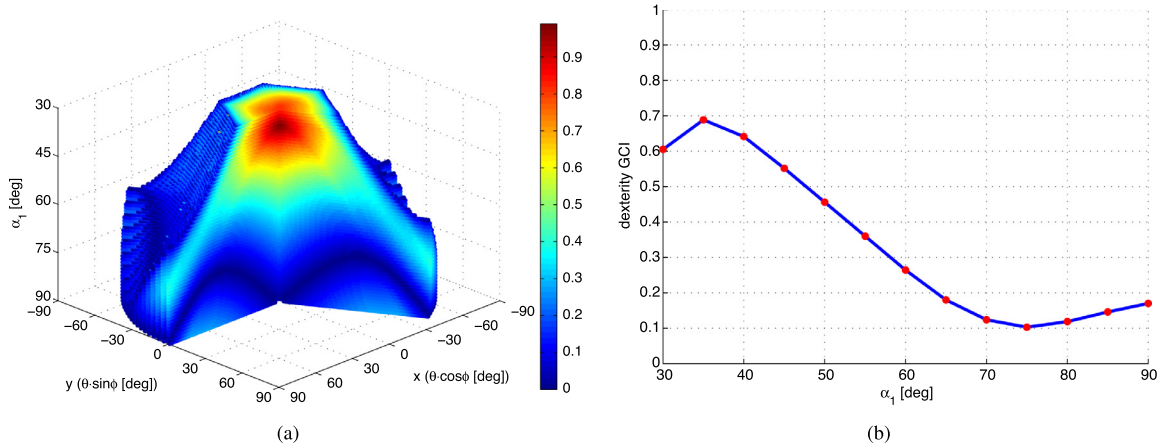


Fig. 7. The overall isocontours of the conditioning index with varying α_1 : (a) isocontours with zero twist; (b) GCI.

The zero-twist reachable workspace, subject to Eqs. (17a)–(17c) with different configurations of $\alpha_1 \in [30^\circ, 90^\circ]$, is shown in Fig. 5. The workspace is bounded by a beautiful tower shape for the given range of α_1 . With $\alpha_1 \in [30^\circ, 60^\circ]$, any slice of the workspace is in the shape of hexagon with arc sides, and it becomes a dodecagon when $\alpha_1 \in (60^\circ, 90^\circ)$ and a circle if $\alpha_1 = 90^\circ$. The larger the angle α_1 is, the larger the reachable workspace is. Moreover, the relationship between the angle α_1 and the minimum tilt angle θ corresponds to the previous findings.

Workspace in polar coordinate system with three configurations of

$\alpha_1 = 45^\circ$, $\alpha_1 = 60^\circ$ and $\alpha_1 = 75^\circ$ is visualized in Fig. 6, respectively. Any σ -twist workspace can be attained by rotating this workspace with angle σ . Thus, a spherical cap of 45° is identified as the regular workspace for full-twist orientation workspace in the configuration of $\alpha_1 = 45^\circ$, together with $\theta \in [0, 60^\circ]$ for $\alpha_1 = 60^\circ$ and $\theta \in [0, 75^\circ]$ for $\alpha_1 = 75^\circ$.

4.2. Kinematic dexterity

Dexterity analysis of a parallel manipulator is directly related to its kinematic Jacobian matrix. Dexterity is defined as the capacity of the

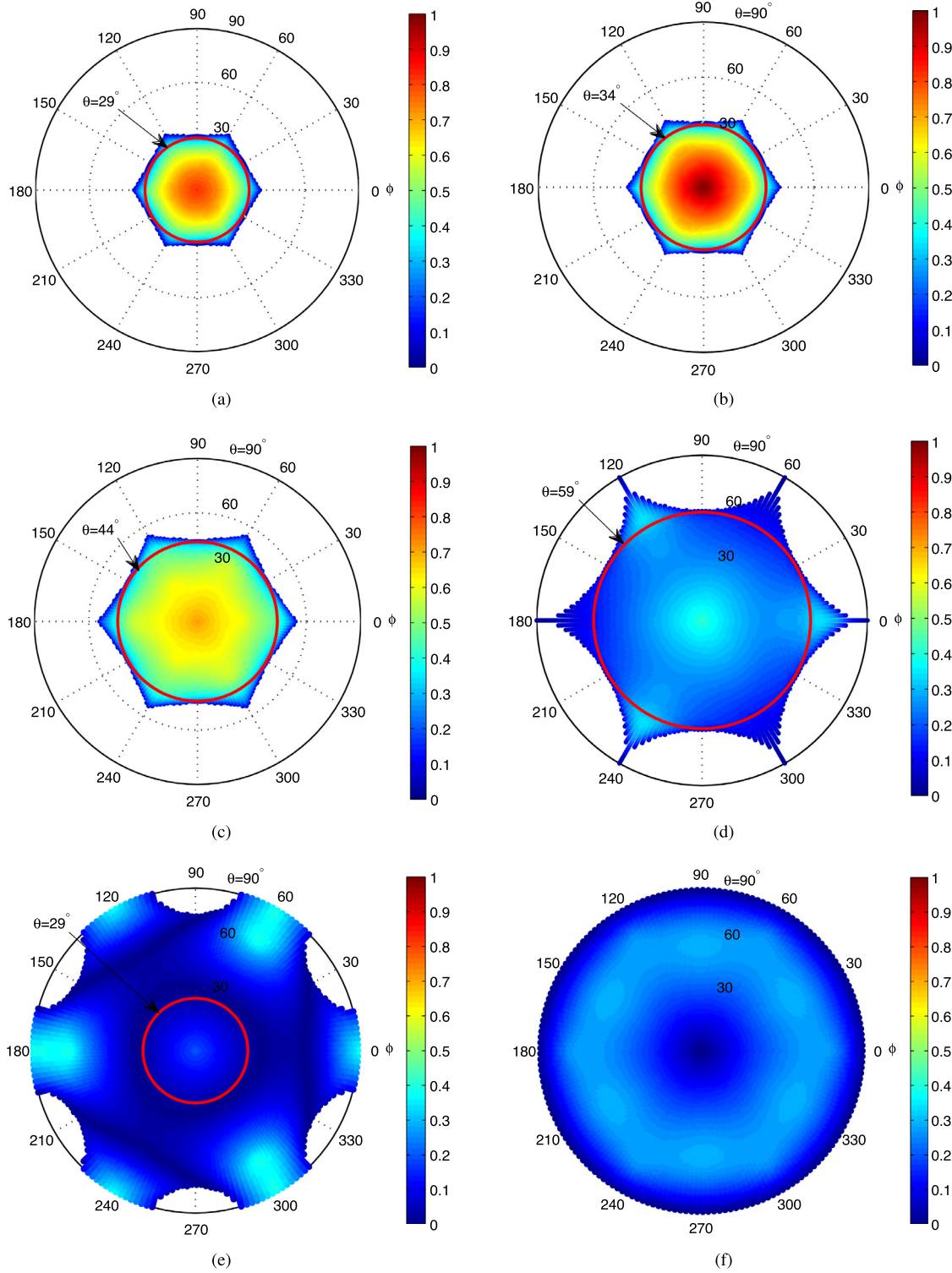


Fig. 8. The isocontours of the conditioning index with zero-twist: (a) $\alpha_1 = 30^\circ$; (b) $\alpha_1 = 35^\circ$; (c) $\alpha_1 = 45^\circ$; (d) $\alpha_1 = 60^\circ$; (e) $\alpha_1 = 75^\circ$; (f) $\alpha_1 = 90^\circ$.

SPM to provide a large range of orientation to its end-effector. The widely used measure index of dexterity is based on the condition number of the Jacobian matrix, defined by

$$\kappa(\mathbf{J}) = \|\mathbf{J}^{-1}\| \|\mathbf{J}\| \quad (20)$$

where $\|\cdot\|$ stands for the Euclidean norm of a matrix. The dexterity index is defined by $1/\kappa(\mathbf{J})$, and the global dexterity index (GCI) [36] can be defined as

$$GCI = \frac{1}{W} \sum_{w_i \in \Omega} \frac{1}{\kappa_i(\mathbf{J})} dw_i = \frac{1}{n} \sum_{i=1}^n \frac{1}{\kappa_i(\mathbf{J})} \quad (21)$$

where $\kappa_i(\mathbf{J})$ is the condition number at a discrete point w_i of the global workspace Ω , and n is the number of the discrete workspace point.

The distributions of the dexterity index of the SPM with varying α_1 over the reachable workspace are depicted in Fig. 7 as well as GCI. It is found that both the maximum local and global conditioning indices appear at $\alpha_1 = 35^\circ$. The regular workplace with $1/\kappa_i(\mathbf{J}) > 0$ becomes

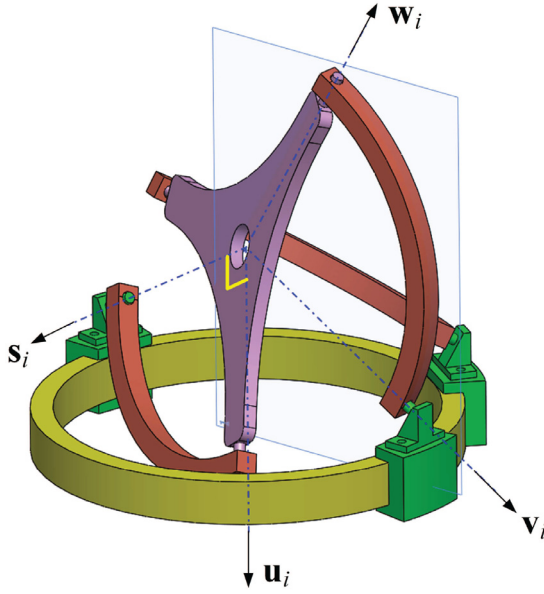


Fig. 9. A type-I singular configuration.

larger with the increasing angle α_1 , when $\alpha_1 \in [30^\circ, 60^\circ]$. Although the regular workspace increases with $\alpha_1 \in (60^\circ, 90^\circ]$, configurations with $\kappa_i(J) = 0$ occur internally, which means the decreased dexterous workspace due to the presence of possible singular configurations. It is noteworthy that the singular configurations exist at the neutral positions when $\alpha_1 = 90^\circ$. Moreover, the GCI index decreases with the increasing α_1 after the peak value at $\alpha_1 = 35^\circ$, and the minimum GCI appears at $\alpha_1 = 75^\circ$.

Fig. 8 shows the isocontours of the conditioning index over the reachable workspace in six configurations, which shows that the SPM has better dexterity in the central region of the workspace and it decreases towards the boundaries except $\alpha_1 = 90^\circ$. When $\alpha_1 = 35^\circ$, the indices are larger within a small workspace, which can reach up to 0.99 maximally. With the increasing angle α_1 , the conditioning index reduces. With $\alpha_1 \in [30^\circ, 60^\circ]$, a regular workspace of a given α_1 with $1/\kappa_i(J) > 0$ is identified as $\theta \in [0, \theta_{\max}]$, $\theta_{\max} = \alpha_1 - 1^\circ$. On the other hand, this regular workspace dramatically decrease with the increasing α_1 in the range of $[60^\circ, 90^\circ]$. It is noteworthy that the dexterity indices with $\alpha_1 = 90^\circ$ decrease to zero at the home configuration, and the dexterous workspace are located in the region of $\theta \in [45^\circ, 75^\circ]$, as shown in Fig. 8(f). Similarly, performance indices with different twist angle σ can be obtained by rotating these figures with the

corresponding σ angle.

4.3. Singularity loci

The manipulator reaches a singularity when the determinant of its forward or/and the determinant of its inverse Jacobian matrix is (are) equal to zero [37], namely,

- the type-I singularity, also named as serial singularity, occurs when $\det(B) = 0$;
- the type-II singularity, also named as parallel singularity, occurs when $\det(A) = 0$.

4.3.1. Type-I singularity

Type-I singularity occurs when the determinant of the inverse Jacobian matrix is equal to 0, namely,

$$\det(B) = \prod_{i=1}^3 \mathbf{u}_i \cdot (\mathbf{v}_i \times \mathbf{w}_i) = \prod_{i=1}^3 \mathbf{u}_i \cdot \mathbf{s}_i = 0 \quad (22)$$

where $\mathbf{s}_i = \mathbf{v}_i \times \mathbf{w}_i$ stands for a vector normal to the plane determined by the vectors $\mathbf{v}_i$ and $\mathbf{w}_i$, as shown in Fig. 9. In other words, vectors $\mathbf{u}_i$, $\mathbf{v}_i$ and $\mathbf{w}_i$ are coplanar, namely, the axis of rotation of the actuated joint is located in the plane that the curved link lies in.

Fig. 10 displays the corresponding singularity loci with the zero twist angle, which shows that the singularity loci are located on the boundaries of the regular and reachable workspace, the regular workspace being defined as circular area corresponding to each angle α_1 . Thus, the singularity loci in the entire orientation workspace will be part of a right circular cone intersected by $\alpha_1 = 30^\circ$ and $\alpha_1 = 90^\circ$, together with three circles of $\theta = 58^\circ$ and $\theta = 90^\circ$, as displayed in Fig. 10(a). When $\alpha_1 = 90^\circ$, all the three limbs encounter serial singularities, which are coincident at $\theta = 90^\circ$.

4.3.2. Type-II singularity

Type-II singularity occurs when the determinant of the forward Jacobian matrix is equal to 0, namely,

$$\det(A) = \begin{vmatrix} (\mathbf{v}_1 \times \mathbf{w}_1)^T \\ (\mathbf{v}_2 \times \mathbf{w}_2)^T \\ (\mathbf{v}_3 \times \mathbf{w}_3)^T \end{vmatrix} = \begin{vmatrix} \mathbf{s}_1^T \\ \mathbf{s}_2^T \\ \mathbf{s}_3^T \end{vmatrix} = \mathbf{s}_1^T (\mathbf{s}_2 \times \mathbf{s}_3) = \mathbf{s}_2^T (\mathbf{s}_3 \times \mathbf{s}_1) = \mathbf{s}_3^T (\mathbf{s}_1 \times \mathbf{s}_2) = 0 \quad (23)$$

Sequentially, two singular configurations as shown in Fig. 11 can be found:

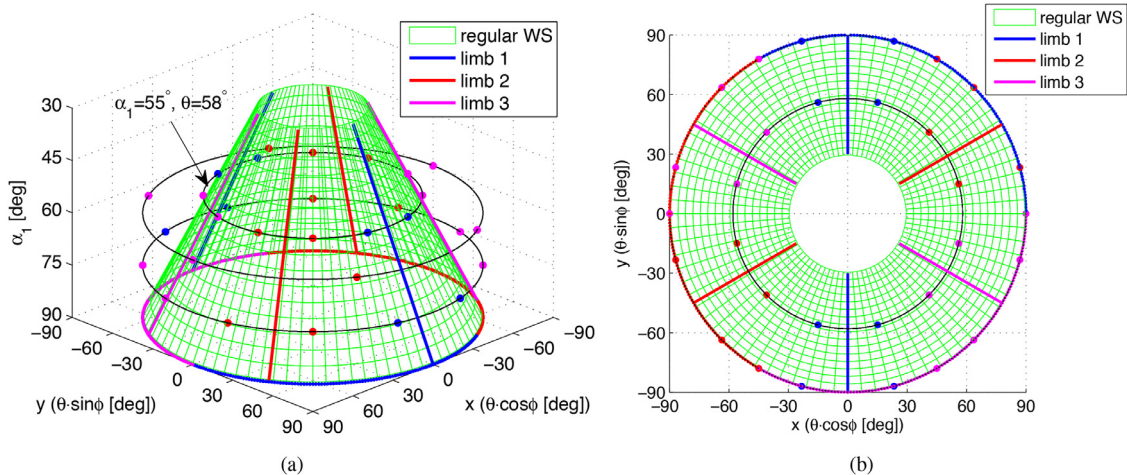


Fig. 10. Type-I singularity loci and the regular workspace with twist angle $\sigma = 0$: (a) isometric view; (b) top view.

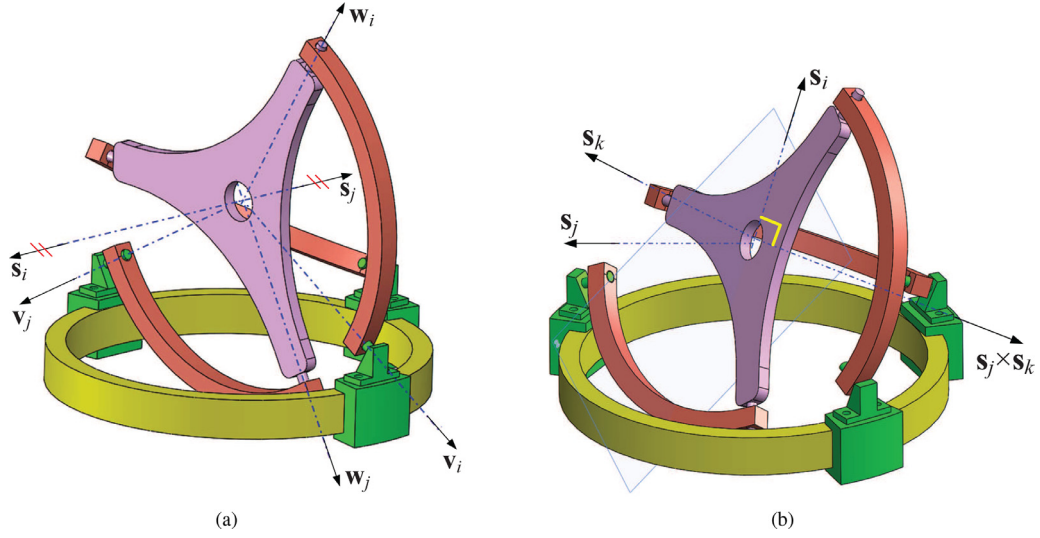


Fig. 11. Type-II singular configurations: (a) constraint singularity; (b) actuation singularity.

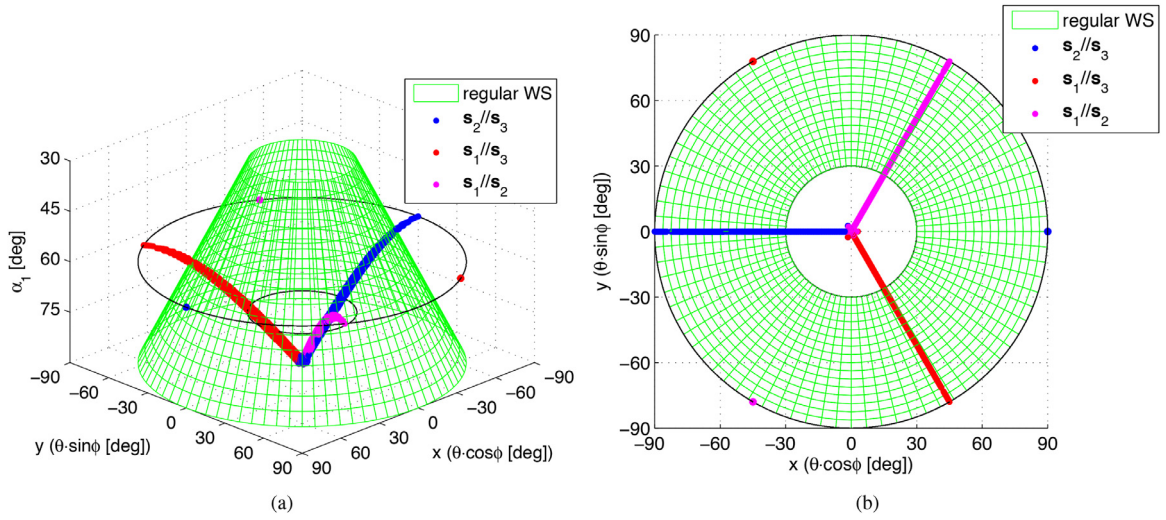


Fig. 12. Constraint singularity loci with $\alpha_1 \in [30^\circ, 90^\circ]$ and the regular workspace with twist angle $\sigma = 0$: (a) isometric view; (b) top view.

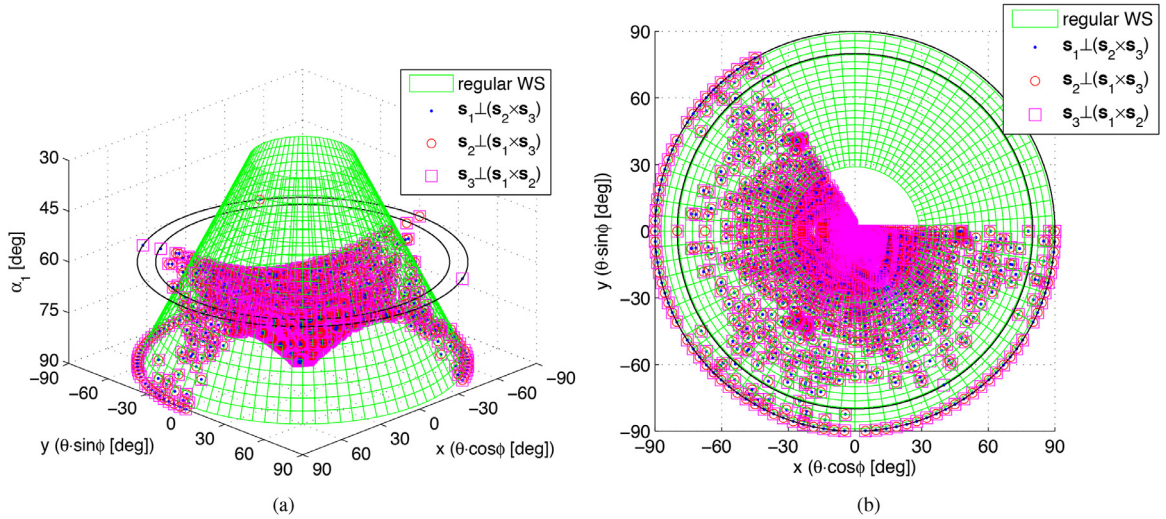


Fig. 13. Actuation singularity loci with $\alpha_1 \in [30^\circ, 90^\circ]$, $\phi \in [2\pi/3, 2\pi]$ and the regular workspace with twist angle $\sigma = 0$: (a) isometric view; (b) top view.

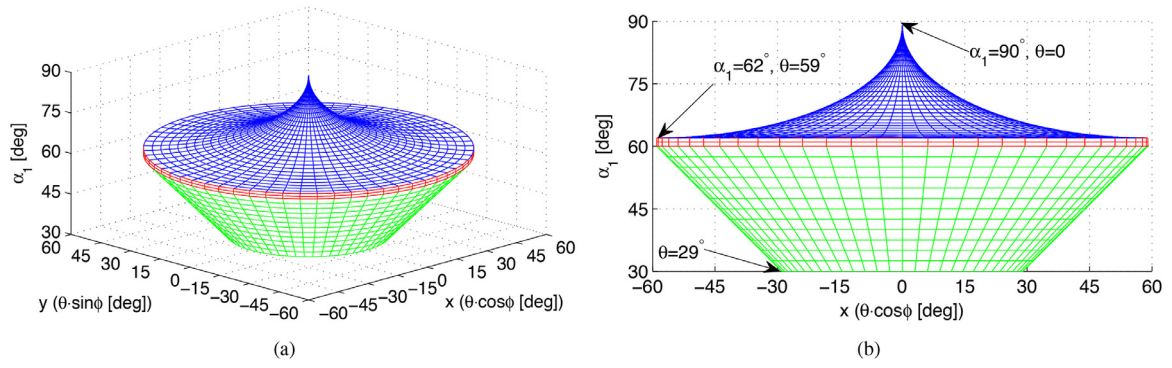


Fig. 14. The full-twist singularity-free workspace with varying α_1 : (a) isometric view; (b) front view.

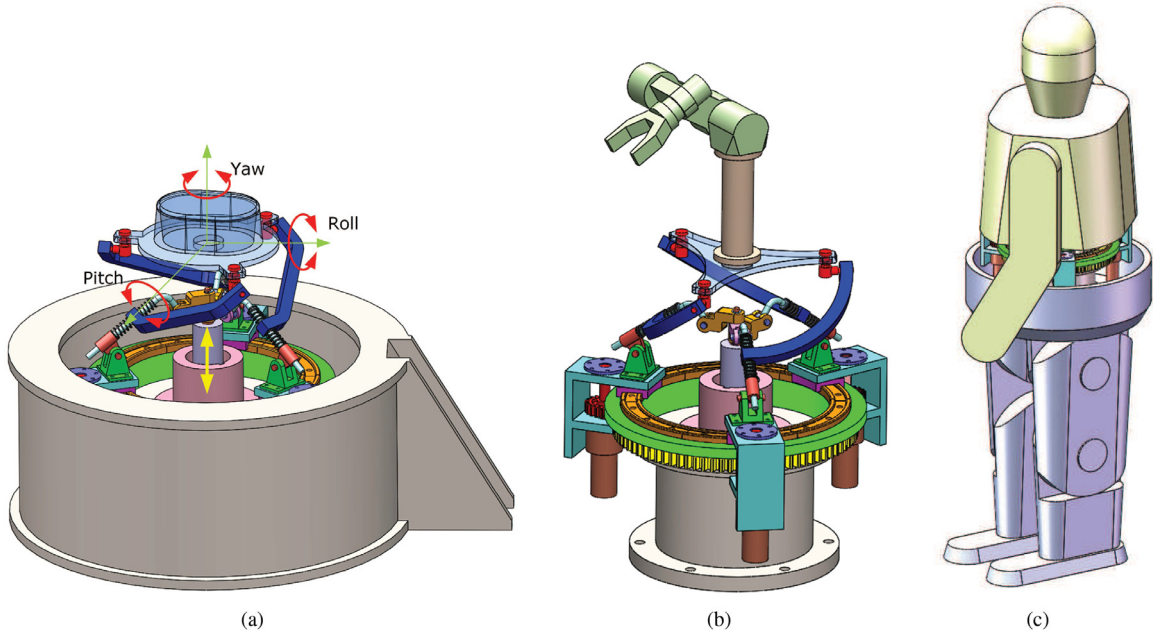


Fig. 15. Applications of the reconfigurable SPM: (a) motion simulator; (b) active spherical joint; (c) waist joint in a humanoid robot.

- Fig. 11(a) shows a singular configuration corresponding to $\mathbf{s}_i$ being parallel to $\mathbf{s}_j$, $i \neq j$, which means that $\mathbf{s}_i \times \mathbf{s}_j = \mathbf{0}$ becomes a zero vector, allowing the manipulator to have additional mobilities even though all the actuated joints are locked, i.e., the constraint singularity [38–42].
- Fig. 11(b) shows the actuation singularity in the case of the general linear complex, namely, $\mathbf{s}_i^T(\mathbf{s}_j \times \mathbf{s}_k) = 0$, $\{i, j, k\} \in \{1, 2, 3\}$, $i \neq j \neq k$. The torque exerting on the moving platform is generated by one of the transmitted torques that is proportional to those that are generated by the other two transmitted torques, thus, the manipulator loses 1-DOF motion in this singular configuration.

The constraint singularity loci are displayed in Fig. 12, which do not exist if $\alpha_1 \in [30^\circ, 60^\circ]$. The constraint singularities occurs in the internal regular workspace, corresponding to the dexterity distributions in Fig. 7(a). Considering the full-circle twist σ , the singularity loci form a hyperboloid.

The actuation singularities with $\phi \in [2\pi/3, 2\pi]$ are depicted in Fig. 13, where the singularity loci are located in and out of the reachable regular workspace. It is noteworthy that the most singular configurations in the three cases are coincident, with a few exceptions. All the singularity loci with full-circle twist form a funnel shape, together with two circles of $\theta = 80^\circ$ and $\theta = 90^\circ$.

Based on the previously analysis, namely, the dexterity distributions and the presence of the singularity loci within the workspace, a

singularity-free workspace shown in Fig. 14 is identified with the varying α_1 , by removing all the singularity loci from the regular workspace as displayed Fig. 10(a). The figure can indicate the operational workspace of the manipulator during its working mode, without passing any singular configuration even during the working mode changes. It is noted that the upper bound of α_1 is 90 degrees, which is indicated by the sharp tip of Fig. 14(b). when α_1 approaches to 90 degrees, the manipulator is close to a singular configuration. From the perspectives of performance and application, the structural design of the SPM can be carried out with an adjustable angle $\alpha_1 \in [30^\circ, 60^\circ]$, since this design admits a relatively larger dexterous workspace.

5. Possible applications of the manipulator

From the perspective of mobilities, this SPM can be considered as a new parallel manipulator with three rotations and one translation (3R1T). Therefore, the proposed manipulator can have two working modes, namely, the 4-dof 3R1T motion and the reconfigurable 3-dof spherical motion. This characteristics ensures some possible applications of the manipulator.

One possible application of the reconfigurable SPM is the motion/flight simulator [43], as shown in Fig. 15(a), thanks to the three rotations of the moving platform as well as the additional vertical translation during its reconfiguration. Compared to the 6-dof Stewart

platform, this manipulator enhances the rotational capability, particularly a full-circle twist. And the decoupled vertical translation eases the control scheme.

Fig. 15(b) shows that the reconfigurable SPM can function as the active spherical joint [44] to replace the actuated serial wrist mechanism for a compact and lightweight design, where a serial robotic arm can be mounted on the moving platform, as shown in Fig. 15(b). Larger tilt angle, namely, larger orientation workspace, can help to extend the reachable region of the end-effector of the hybrid arm manipulator, and the enhanced dexterity via the reconfiguration can ensure the agile motion of the arm or the high motion accuracy.

Fig. 15(c) shows another potential application of the SPM as a waist joint in legged robots, for example, a humanoid robot, which can behave different kinematic performances in different configurations. Moreover, it can realize the unidirectional bending for the upper body due to the full-circle twist motion. The adoption of the waist joint in the humanoid can help keep balance during walking, thus improve the stability of walking particularly on uneven terrains [45,46]. Moreover, the waist joint enables the humanoid for more movement at the waist so it can move in a more natural way similar to human beings.

6. Conclusions

In this paper, the 3-RRR spherical parallel manipulator of co-axis input, featuring a full-circle twist movement, is redesigned with reconfigurability to extend the applications of the wrist mechanisms. The reconfiguration approach is to adopt a four-bar linkage in each independent limb, with synchronized motion in all the three limbs, which can be readily built thanks to its simple structure. The variations of the manipulator performance can be achieved by adjusting one geometric parameter, to ease the control. Moreover, the new design can be used directly for applications such as flight simulator or robotic joints.

The reconfiguration strategy can either allow the manipulator to work in a prescribed configuration of 3-dof rotational movements or ensure the changed orientational workspace size and kinematic dexterity during changing configurations, namely, online reconfiguration. Although the manipulator singularities exist in any configuration corresponding to the adjustable geometric angle, a singularity-free workspace is identified for avoidance. Moreover, from the perspective of mobilities, the proposed manipulator can have two working modes, namely, the 4-dof 3R1T motion and the reconfigurable 3-dof spherical motion.

Some new features are revealed for the reconfigured SPM. Its zero-twist workspace is found to be a nice shape of tower. Moreover, a linear influence of the adjustable angle in the range of $[30^\circ, 60^\circ]$ on the regular workspace size is revealed, namely, the increased angle leading to an increasing singularity-free workspace size but lower dexterity. Singularity loci are identified to detect a singularity-free workspace with the varying α_1 , which can indicate the operational workspace of the manipulator during its working mode, without passing any singular configuration. This reconfigurable spherical parallel manipulator can meet various task requirements for its flexibility and multiple working modes.

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